INCH-POUND

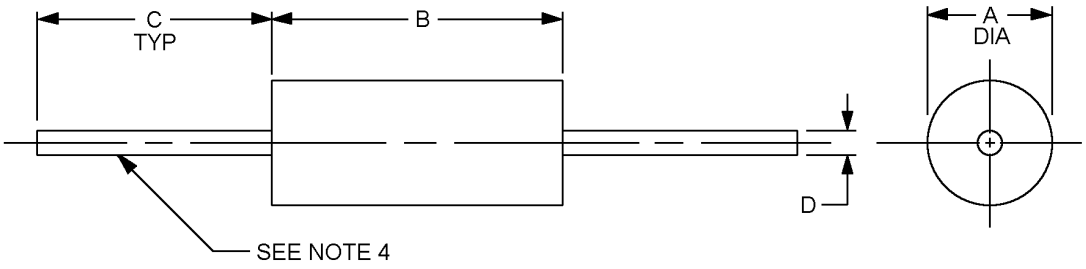
MS21427C
W/AMENDMENT 1
15 July 2022
SUPERSEDING
MS21427C
24 August 2007

MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, MICRO-MINIATURE, MAGNETICALLY SHIELDED, (FERRITE CORE – FERRITE SLEEVE), TYPE LT10K

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	Dimensions in inches with metric equivalents (mm) in parentheses
	Minimum	Maximum
A	.085 (2.16)	.105 (2.67)
B	.240 (6.10)	.260 (6.60)
C	1.380 (35.05)	1.620 (41.15)
D	.015 (0.38)	.0215 (0.55)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their leads.
4. Solderable/weldable lead wire, AWG number 26.

FIGURE 1. Dimensions and configuration.

AMSC N/A
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FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT10K

Grade: 1
Class: A

Weight: .250 grams, maximum.

Operating temperature range: -55° to +105°C.

Ambient temperature: +90°C maximum.

Temperature rise: 15°C.

Terminal pull: 3 pounds minimum.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 200 V rms.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 80 V rms.

Percent coupling: 3 percent, maximum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I. DC resistance shall be the last measurement taken in the electrical characteristics test sequence.

Part or Identifying Number (PIN): MS21427 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

Dash Number MS21427	Type designation	Inductance (μH) ±10%	Q (min)	Test frequency (MHz)	SRF min (MHz)	DC resistance at 25°C max (ohms)	Rated DC current (mA)	Incremental current (mA)
-1	LT10K555	120	31	.79	13	5.8	88	27
-2	LT10K556	150	33	.79	12	7.9	75	24
-3	LT10K557	180	33	.79	11	9.4	69	22
-4	LT10K558	220	35	.79	10	11.0	64	20
-5	LT10K559	270	37	.79	9	12.0	61	18
-6	LT10K560	330	40	.79	8	16.0	53	16
-7	LT10K561	390	38	.79	7.8	21.0	46	14
-8	LT10K562	470	36	.79	7.5	24.0	43	13
-9	LT10K563	560	36	.79	7.0	28.0	40	12

1/ The dash number added to the MS military standard number constitutes the MS part number, for example MS21427-1

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement	Allowable variation from Initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5	---	---	-15
Group III	±10	±(3% +.001 ohm)	-15	-20
Group IV	±10	±(3% +.001 ohm)	-15	-20
Conformance inspection group C				
Subgroup I	±5	---	---	-15
Subgroup II	±10	±(3% +.001 ohm)	-15	-20
Subgroup III	±10	±(3% +.001 ohm)	-15	-20

NOTES:
1. The SRF minimum values tabulated exceeding 250 MHz are estimates and to be used for reference only.
2. Polarization during moisture resistance test is not applicable.
3. Terminal strength twist test, in accordance with MIL-STD-202, method 211, test condition D, except 180° rotation for a total of 540°.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Air Force – 19

Preparing activity:
DLA – CC

(Project 5950-2022-055)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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